

Hierarchical autoregressive neural networks in three-dimensional statistical system

Mateusz "■" Winiarski

under supervision of dr Tomasz Stebel

experimental physics[†], WFAIS UJ

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Neural network: what you think it is

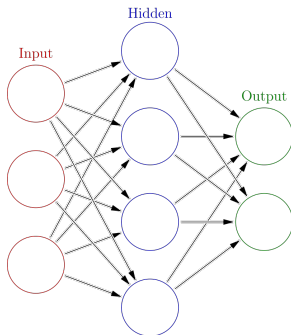


Figure: First Google Image search result for string "neural network"



Neural network: what you think it is

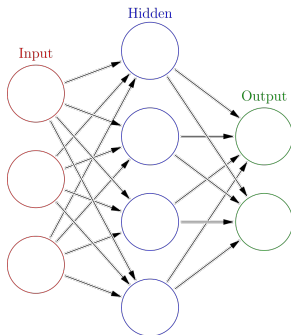


Figure: Your reaction to that information

Figure: First Google Image search result for string "neural network"





Neural network: what it really is

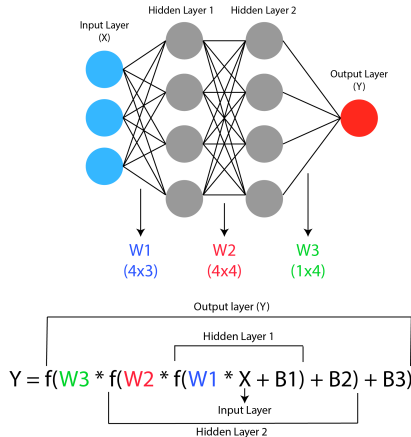


Figure: In physics, everything is a matrix



Autoregression

$$X_t = \sum_{i=1}^p \varphi_i X_{t-i} + \varepsilon_t$$

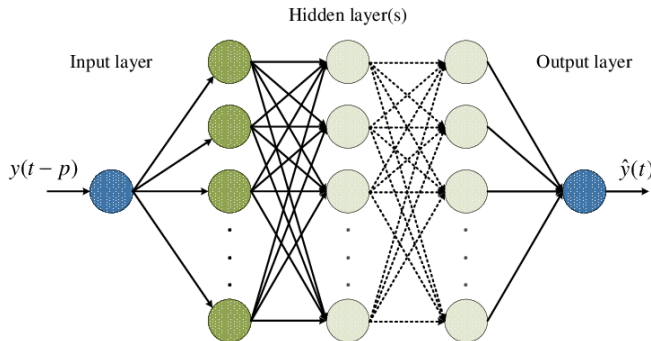


Figure: An example of autoregressive neural network (source: [6])





Hierarchy

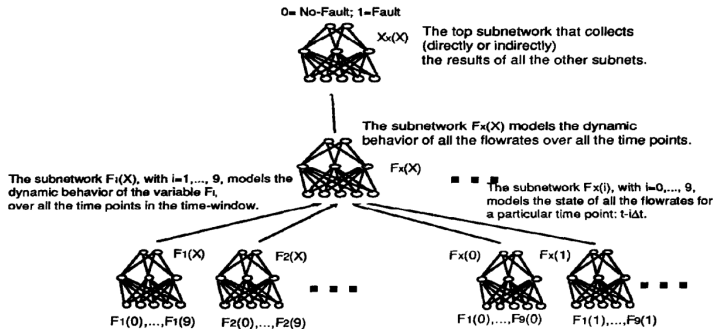
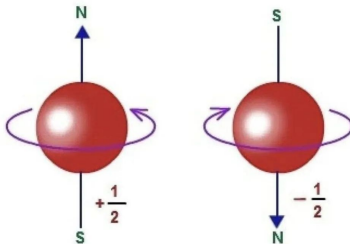


Figure: An example of hierarchical neural network (source: [5])



What is spin

Electron spin explained: imagine a ball that's rotating, except it's not a ball and it's not rotating



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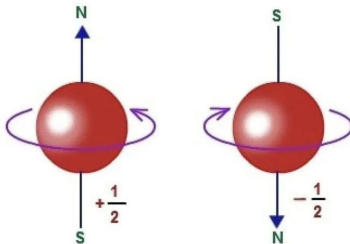


Figure: This is not what spin means



What is spin, again

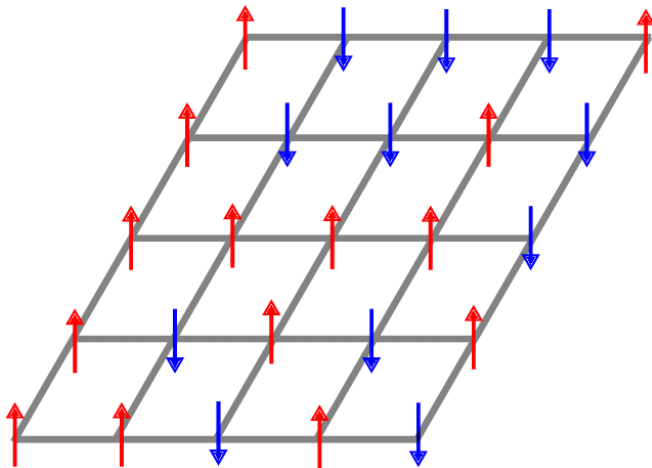


Figure: Illustration of 2D Ising model



Some physics now: Ising model

- ▶ σ – configuration (microstate)
- ▶ assumption: no external magnetic field

$$M(\sigma) = \sum_{i \in \text{spins}} s_i$$

$$E(\sigma) = -\frac{1}{2} \sum_{i,j \in \text{neighbours}} s_i s_j$$

$$p(\sigma) = \frac{e^{-\beta E(\sigma)}}{Z}$$

$$Z = \sum_{\sigma} e^{\beta E(\sigma)}$$

$$F = -\frac{1}{\beta} \log Z$$

$$S = \beta(U - F)$$

$$\beta = \frac{1}{k_B T}$$

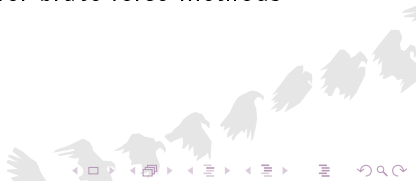


Limitations of MCMC algorithms

(MCMC – Monte Carlo Markov chain)

Limitations:

- ▶ no access to partition function $Z \rightarrow$ can't compute F and S
- ▶ critical slowing down near phase transitions (at β_c)
- ▶ exponential scaling in 3D systems for brute-force methods



How to apply hierarchy

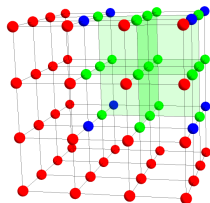


Figure: Hierarchy in 4x4x4 cube

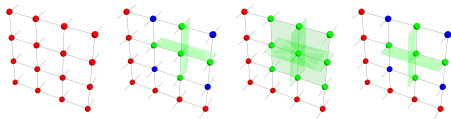


Figure: Slices of the left figure

$$\begin{aligned}
 p(s) &= p(B(s))p(I(s)|B(s)) \\
 &= p(B(s)) \prod_{a=1}^6 p(I^a(s)|B^a(s)).
 \end{aligned}$$



How to apply autoregression

$$F = -\frac{1}{\beta} \log Z$$

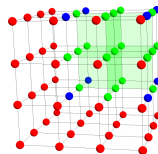
$$F = \frac{1}{\beta} \sum_{i \in \text{configurations}} q_{\theta}(\sigma_i) [\beta E(\sigma_i) + \log q_{\theta}(\sigma_i)]$$



Algorithm workflow

step 1: hierarchical training

- ▶ **level 1**: boundary spins (2D-like network)
- ▶ **level 2**: cross-structure (smaller network)
- ▶ **level 3**: isolated spins (heatbath)



step 2: autoregressive sampling

$$q_{\theta}(s) = \prod_{i=1}^N q_{\theta}(s^i | s^1, \dots, s^{i-1})$$

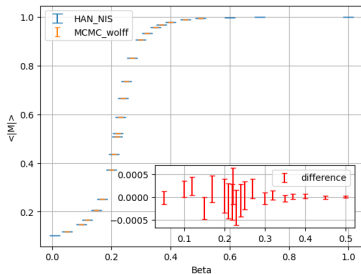
step 3: Neural Importance Sampling (NIS)

$$\langle O \rangle = \frac{1}{N\hat{Z}} \sum_i \hat{v}(s_i) O(s_i)$$

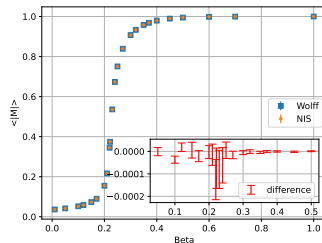


Some plots

Magnetization: Monte Carlo vs neural network



(a) 4x4x4

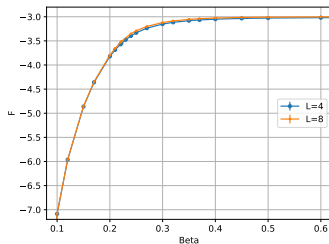


(b) 8x8x8

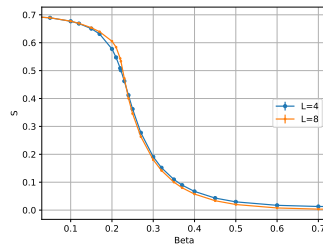
Figure: Magnetization vs temperature function β 

Some plots

Extensive properties: neural network



(a) free energy



(b) entropy

Figure: Extensive parameters vs temperature parameter β for 4x4x4 and 8x8x8 nets



Timing

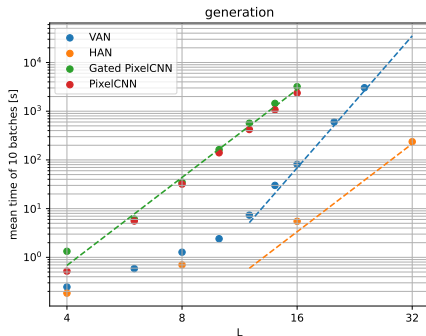
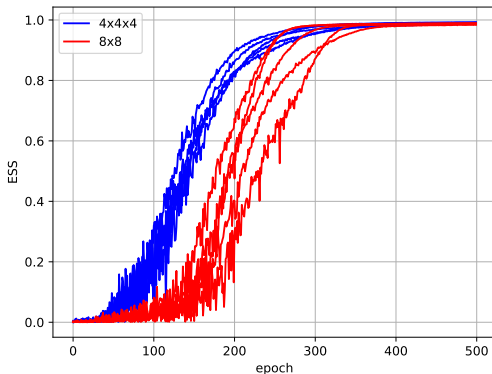


Figure: Mean time (in seconds) of 10 batches generation of VAN (blue circles), HAN (orange circles), PixelCNN (red circles) and Gated PixelCNN (green circles) algorithms in dependence on system linear size L . Asymptotic approximations αL^9 (blue dashed line), βL^6 (orange dashed line) and γL^6 (green dashed line), for VAN, HAN and Gated PixelCNN accordingly, were superposed on plots



Numerical results

Timing: 2D vs 3D



ESS – effective sample size

$$\begin{aligned}
 ESS &= \frac{\langle \hat{w} \rangle_{q_\theta}^2}{\langle \hat{w}^2 \rangle_{q_\theta}} \\
 &\approx \frac{\left(\sum_{i=1}^N \hat{w}(s_i) \right)^2}{N \sum_{i=1}^N \hat{w}^2(s_i)}
 \end{aligned}$$



Figure: Comparison of the ESS growth for 4x4x4 (blue) and 8x8 (red) systems in a corresponding critical temperature β (0.22 for the 3D and 0.38 for the 2D system)

Timing: comparison to classical algorithms

parameters: 10^9 configurations, magnetization error $2 \cdot 10^{-6}$

- ▶ HAN: 430 s for training + 3400 s for generation
- ▶ Wolff algorithm: 47 s



Bibliography

- [1] Piotr Białaś, Vaibhav Chahar, Piotr Korcyl, Tomasz Stebel, Mateusz Winiarski, and Dawid Zapolski. “Hierarchical autoregressive neural networks in three-dimensional statistical system”. In: *arXiv preprint arXiv:2503.08610* (2025). URL: <https://arxiv.org/abs/2503.08610>.
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- [6] Mihail Balanici and Stephan Pachnicke. “Server Traffic Prediction Using Machine Learning for Optical Circuit Switching Scheduling”. In: May 2019.



Thanks for attention



<https://github.com/falcotinnunculus/przezrocza/blob/wladca/semmag/main.pdf>

